

# Vijaya Ragunath S

AI / ML Engineer

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## Summary

M.Sc. Data Science graduate and AI/ML intern with hands-on experience building LLM-powered applications in Python and FastAPI using local and cloud LLMs (Ollama, Google Gemini). Practical exposure to RAG, embeddings, vector databases, and prompt engineering, on a foundation in machine learning and deep learning.

## Experience

### VDart

Nov 2025 – Mar 2026

#### AI/ML Intern

- Worked within an engineering team building AI features, learning how AI applications are developed and how backend services communicate with AI models in a professional environment.
- Built FastAPI backend endpoints connecting application logic to large language models, working with both a cloud API (Google Gemini) and locally hosted models via Ollama.
- Gained hands-on experience with core LLM patterns — Retrieval-Augmented Generation, embeddings, vector databases, and prompt engineering — through guided implementation work.
- Experimented with chatbot and document-retrieval functionality to understand how retrieval, embeddings, and LLM synthesis combine into a working system.

### VDart

May 2025 – Jun 2025

#### Data Analyst Intern

- Built interactive Power BI dashboards with drill-down and filtering to support reporting.
- Cleaned and transformed datasets using SQL and Excel to prepare data for analysis.
- Created DAX measures (CALCULATE, RANKX, ALL) to compute and analyze business metrics.

## Projects

### AI English Coach | Python, Streamlit, Ollama (Mistral 7B), OpenAI Whisper, gTTS

- Built an AI-powered English speaking-practice app with voice and text input, structured grammar correction, and an LLM-generated fluency score, across Practice and Interview modes and three difficulty levels.
- Integrated a locally hosted LLM (Mistral 7B via Ollama) with a structured prompt returning a corrected sentence, natural response, improvement tip, and practice question — no external API dependency.
- Implemented a full voice loop — speech-to-text with OpenAI Whisper and spoken replies via gTTS — with session-state conversation tracking, a progress dashboard, and JSON session-report export.
- Added error handling with user-facing fallbacks around the LLM, speech-to-text, and text-to-speech calls, and cached the Whisper model load to avoid reloading it on every interaction.

### Document Q&A with RAG | Python, FastAPI, ChromaDB, Embeddings, Ollama (Mistral 7B), Google Gemini

- Built a Retrieval-Augmented Generation app where users upload PDFs and ask questions answered from the document content rather than the LLM's general knowledge.
- Implemented the full pipeline in Python and FastAPI: PDF text extraction, chunking, embedding generation, storage and similarity search in ChromaDB, and retrieval of the most relevant chunks per question.
- Injected retrieved context into the prompt and generated answers using a local LLM (Mistral 7B via Ollama) and Google Gemini.
- Exposed the pipeline through FastAPI endpoints and tested the full document upload → retrieval → answer-generation workflow locally, end to end.

## Technical Skills

- **Languages:** Python, SQL, JavaScript, TypeScript, HTML, CSS
- **Machine Learning:** Regression, Classification, Decision Trees, Random Forest, Gradient Boosting, KNN, SVM, Feature Engineering, Model Evaluation
- **Deep Learning:** ANN, CNN, RNN, LSTM, TensorFlow, Keras
- **Generative AI / LLMs:** Prompt Engineering, RAG, Embeddings, Vector Databases, LLM Integration (Ollama, Google Gemini), Whisper (STT), gTTS (TTS)
- **Backend & Web:** FastAPI, REST APIs, React, Streamlit
- **Databases & Tools:** PostgreSQL, SQLite, ChromaDB, Git, GitHub, Docker (basic), Postman, Power BI, DAX

## Education

### SRM Institute of Science and Technology

2024 – 2026

M.Sc. Data Science

### SRM Trichy Arts and Science College

2020 – 2023

B.Sc. Computer Science

## Certifications

- Microsoft Certified: Power BI Data Analyst Associate (PL-300)
- LinkedIn Learning — Python for Machine Learning
- LinkedIn Learning — Applied Machine Learning